CASE STUDY: Design and construction of an autonomous robot with room-mapping capabilities.

Project objective.

The objective of this project is to design and build a robot capable of navigating a room autonomously, mapping its layout into its memory, and subsequently visualizing the space on a simple 2D grid.

Concept and algorithms.

After analyzing existing market solutions and hobbyist projects, we decided to implement the SLAM (Simultaneous Localization and Mapping) algorithm. This process involves simultaneously determining the robot's location and mapping the walls within its memory. To generate an accurate map of a room, the robot must know its precise position—specifically, its coordinates and its orientation (angle) relative to the starting point. The simplified operational workflow of our robot is as follows:

1. Position tracking – the robot calculates the position of its center based on data regarding wheel movement.

2. Environmental scanning – after traveling a set distance, the robot stops to scan its surroundings. To achieve this, we will use LiDAR (Light Detection and Ranging) technology, which employs a laser that reflects off surfaces in the robot's environment to measure distances.

3. Data integration – the robot combines its positional data with the information gathered via the LiDAR system. Using mathematical calculations, it determines the relevant angles and distances.

4. Map generation – based on the calculated distances and angles, the system records the data onto a grid.

Selection of the main microcontroller. To implement the SLAM algorithm, a microcontroller must have sufficient memory and the capability to perform rapid trigonometric calculations.

1. Raspberry Pi 4 – This was our first choice, as we already have one. However, it is not strictly a microcontroller, but rather a miniature computer. Notably, there is a specialized framework for robotics called ROS, which is compatible with the Raspberry Pi.

2. Arduino UNO (ATmega328PB) – Insufficient memory!

3. ESP32-S3 – We selected this microcontroller because it has enough memory. Additionally, it includes built-in Wi-Fi and Bluetooth modules.

Selection of motors and a system for tracking the robot's position.

In robotics, this is technically referred to as "odometry"—the process of determining the distance traveled and the robot's orientation. Without knowing exactly how many degrees or millimeters the robot has moved, it is impossible to accurately map its position onto a grid. Our analysis:

1. TT motors (yellow) with optical encoders – These are generally not recommended for projects of this type, as they can lead to inaccurate position tracking. Nevertheless, they may be suitable for low-budget projects. We have not entirely ruled them out and might still use them.

2. Stepper motors – These offer high measurement repeatability and are easy to use. Unfortunately, they consume a significant amount of power and are relatively heavy.

3. DC motors with quadrature encoders, e.g., the JGA25-370: these provide precise readings, though they require more code to implement.

Sensors for wall scanning:

1. Ultrasonic sensor, e.g., the HC-SR04 – sound waves are poorly suited for room mapping because they propagate in a cone shape rather than in straight lines. Furthermore, they are prone to reflections, which can interfere with readings. They are best used simply for distance measurement in less complex projects.

2. Off-the-shelf 360-degree LIDAR scanner – the most accurate solution; however, it exceeds our budget, as such units cost upwards of 300 PLN.

3. Custom-built 2D LIDAR – we intend to construct our own scanner. It will consist of a servo motor (e.g., an SG90) and a Time-of-Flight (ToF) sensor (e.g., a VL53L0X). The servo will perform a slow sweeping motion across a 0–180 degree range, while the sensor measures the distance.

Power supply:

To use the JGA25-370 DC motors, we need a power source capable of providing at least 6V. We plan to use two 18650 lithium-ion cells connected in series, which will yield 7.4V. To properly power each component, we will need a step-down converter to provide 5V (for the servo and the ESP32), while the 3.3V supply will be drawn directly from the ESP32 (via its 3.3V pin).

Camera:

We plan to install an additional ESP32-CAM module on the front of the robot. This module will serve as an independent vision system, providing a live, wireless video feed. It will allow for the observation of the route from the perspective of the robot's bumper.

List of features:

1. Room mapping using ToF (Time-of-Flight) sensors
2. A grid to visualize the placement of obstacles within the room
3. Odometry: calculating the robot's position relative to its starting point and tracking wheel rotations
4. Autonomy: detecting and avoiding obstacles
5. Visualization on the 2D grid, with the generated map transmitted wirelessly to a web browser

Additional features:

1. Live video streaming from an onboard camera